Creality V4.2.2 FAN (PA0) → 5 V TTL adapter (non-inverting)

Fan ON (FAN− ≈ 0 V) → OUT ≈ 5 V | Fan OFF (FAN− ≈ 24 V) → OUT ≈ 0 V | laser disconnected for bench

K-FAN1 or K-FAN2 (shared PA0)

FAN+ (fixed ≈ 24 V)

FAN− (switched drain)

Board / PSU DC− = common GND

Rload 2.2 kΩ 1 W

across FAN+ → FAN−

I≈10.9 mA P≈0.26 W @ 24 V ON

Optional Dfly (UF4007)
cathode→FAN+ if fan remains

Do NOT invent K-FAN pin order:
verify +/− from silk / meter

Sense + invert (2N3904)

FAN− ── R1 47 kΩ ──●── R2 10 kΩ ── GND

|

Rb 4.7 kΩ

|

B of Q1 (NPN)

Q1: 2N3904 / BC547

E → GND C → OUT node

When FAN−≈24 V: Q1 ON → OUT low

When FAN−≈0 V: Q1 OFF → OUT high

(= fan ON → laser ON; MCU-non-inv.)

5 V logic rail + output

Preferred Vcc: external buck 24 V→5 V

from PSU/FAN+ and DC− (common GND)

Documented on-board 5 V also exists on
BL_T / probe and Display EXP — both
OCCUPIED; pin order NOT asserted here.

Rc 1.0 kΩ: Vcc(5 V) → OUT / Q1-C

Rpd 10 kΩ: OUT → GND (fail-safe LOW)

Rs 220 Ω series to laser TTL (optional)

Optional: Cbp 100 nF on Vcc–GND

3.3 V option: same parts, Vcc=3.3 V

(Rc may stay 1 kΩ; levels ~0 / 3.3 V)

Safety / polarity

Laser DISCONNECTED for resistor-load bench. Heaters DISCONNECTED. Scope ground ONLY on board/PSU DC− — never on FAN−.

Unpowered board, missing 5 V, or fan OFF → OUT held LOW by Rpd (and/or Q1 ON when FAN− floats high via Rload).

Common ground: 24 V DC−, adapter GND, 5 V buck GND, laser module GND. No M3 / laser fire without Michael watching.

Why Rload: SCOPE8 saw ~1 kHz duty ripple stuck near 24 V — consistent with probing an always-hot node and/or an unloaded/inductive FAN− that never pulled cleanly. Rload from FAN+ to FAN− forces a defined pull-UP on the drain so OFF≈24 V and ON≈0 V at FAN− vs DC− (tests MOSFET path independent of fan electronics).

Generated by generate_diagrams.py — values match README.md / BOM.csv